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Homework 9

1.

Exercise 4.9 Show that

$$\int_{-\infty}^{\infty} \exp\{-x^2\} \exp\{-(x-\mu)^2 y/2\sigma^2\} \, \mathrm{d}x = \exp\left\{-\frac{\mu^2}{1+2\sigma^2/y}\right\}$$

by completing the square in the exponent to evaluate the integral.

2.

Exercise 4.10 Consider an Accept–Reject method for the target density f based on the instrumental distribution g.

a. Show that the Accept–Reject sample $(X_1, \ldots, X_n)$ can be associated with two iid samples, $(U_1, \ldots, U_N)$ and $(Y_1, \ldots, Y_N)$, with corresponding distributions $\mathcal{U}_{[0,1]}$ and g; N is the stopping time associated with the acceptance of n variables Y_j.
b. Deduce that the corresponding estimator of $\mathbb{E}_f[h(X)]$ based on $(X_1, \ldots, X_n)$ can be written as

$$\delta_1 = \frac{1}{n} \sum_{i=1}^{n} h(X_i) = \frac{1}{n} \sum_{j=1}^{N} h(Y_j) \, \mathbb{I}_{U_j \leq w_j} \, ,$$

with $w_j = f(Y_j)/Mg(Y_j)$.

(a)

(b)

3.

Exercise 4.15 For each of the following cases, generate random variables X_i and Y_i and compare the empirical average and Rao–Blackwellized estimator of $\mathbb{E}_f(X)$ and $\mathrm{var}_f(X)$:

a. $X|y \sim \mathcal{P}(y)$, $Y \sim \mathcal{G}a(a,b)$ (X is negative binomial);
b. $X|y \sim \mathcal{N}(0,y)$, $Y \sim \mathcal{G}a(a,b)$ (X is a generalized t);
c. $X|y \sim \mathcal{B}in(y)$, $Y \sim \mathcal{B}e(a,b)$ (X is beta-binomial).

(b)

(c)

4.

Exercise 4.18 A naïve way to implement the antithetic variable scheme is to use both U and $(1-U)$ in an inversion simulation. Examine empirically whether this method leads to variance reduction for the distributions (i) $f_1(x) = 1/\pi(1+x^2)$, (ii) $f_2(x) = \frac{1}{2}e^{-|x|}$, (iii) $f_3(x) = e^{-x}\mathbb{I}_{x>0}$, (iv) $f_4(x) = \frac{2}{\pi\sqrt{3}}\left(1+x^2/3\right)^{-2}$, and (v) $f_5(x) = 2x^{-3}\mathbb{I}_{x>1}$. Examine variance reductions of the mean, second moment, median, and 75th percentile.

To calculate the weights for the Rao–Blackwellized estimator of Section 4.6, it is necessary to derive properties of the distribution of the random variables in the Accept–Reject algorithm. The following problem is a rather straightforward exercise in distribution theory and is only made complicated by the stopping rule of the Accept–Reject algorithm.

(i)

(ii)

(iv)

5.

Assuming that we use antithetic variables to estimate some parameter of the standard normal distribution, we must prove that the covariance between X_i and Y_i is -1.